

AI Usage Specification

This document explicitly details two distinct dimensions of AI within the AI Study Companion:

- 1. **AI Used by the Product** (Runtime AI capabilities exposed to users)
- 2. **AI Used to Build the Product** (Agentic tooling and prompts employed during engineering)

Part 1: AI Used by the Product (Runtime Systems)

The application employs a provider-agnostic, multi-model architecture. For prototype development and free-tier operation, the default runtime is **Google Gemini (free tier)**, while retaining clean swappability to Anthropic Claude or OpenAI via the unified `LLMProvider` / `EmbeddingProvider` abstraction layer.

Feature / Subsystem	Default Provider & Model	Fallback / Alternative	Role & Implementation	Temperature / Config	Output Validation
Semantic Embedding	Google Gemini <code>text-embedding-004</code>	OpenAI <code>text-embedding-3-small</code> / <code>@xenova/transformers</code>	Converts document chunks and learner queries into 768-dimensional dense vectors for cosine similarity search in <code>pgvector</code> .	N/A	Strict 768 float array validation
Concept Extraction	Google Gemini <code>gemini-2.0-flash</code>	Anthropic <code>claude-sonnet-4-20250514</code>	Reads initial material text chunks in background worker to distill 5–15 atomic learning concepts per project.	0.2, <code>maxTokens</code> 1000	Structured JSON schema (<code>{ concepts: [{ name, description }] }</code>)
Grounded AI Tutor	Google Gemini <code>gemini-2.0-flash</code>	Anthropic <code>claude-sonnet-4-20250514</code>	Conversational Socratic tutor streaming grounded explanations with citations <code>[Material Name, Page X]</code> and strict refusal when	0.4, <code>maxTokens</code> 2000	Streaming SSE, citation metadata extraction, prompt injection defense

Feature / Subsystem	Default Provider & Model	Fallback / Alternative	Role & Implementation	Temperature / Config	Output Validation
			evidence is absent.		
Adaptive Quiz Generation	Google Gemini <code>gemini-2.0-flash</code>	Anthropic <code>claude-sonnet-4-20250514</code>	Generates mixed multiple choice (MCQ) and open-ended conceptual questions targeted at the learner's weakest concepts.	0.5, maxTokens 1200	Strict Zod schema validation (4 options for MCQ, explicit correct choice, explanation)
Open-Ended Quiz Grading	Google Gemini <code>gemini-2.0-flash</code>	Anthropic <code>claude-sonnet-4-20250514</code>	Evaluates learner's written explanation against model answers and evaluation rubrics.	0.1, maxTokens 600	Structured JSON (<code>{ score: 0-100, isCorrect: boolean, feedback: string, missingConcepts: string[] }</code>)
Personalized Recommendations	Google Gemini <code>gemini-2.0-flash</code>	Anthropic <code>claude-sonnet-4-20250514</code>	Analyzes learner mastery distribution, recent mistake patterns, and goal text to output 2-3 actionable next steps.	0.3, maxTokens 800	Structured JSON (<code>{ recommendations: [{ text, reason, priority }] }</code>)

Observability & Guardrails

- **AIUsageLog** : Every runtime AI call records `promptTokens` , `completionTokens` , `latencyMs` , `costUsd` , `model` , `feature` , `userId` , `projectId` , and `success` .
- **System Prompts**: System prompts explicitly enforce boundary sandboxing, source citing, and refusal when facts are not in the document context.
- **Rate Limiting**: AI endpoints enforce sliding-window token bucket limits to prevent denial-of-wallet attacks.

Part 2: AI Used to Build the Product (Engineering Workflow)

During the development of the AI Study Companion, autonomous AI coding agents (DeepMind Antigravity / Gemini) were utilized across the entire software development lifecycle:

1. System & Schema Design:

- Agent analyzed the PRD and drafted the complete 18-model Prisma schema with relational constraints, enums, cascades, and vector definitions.

2. Full-Stack Implementation:

- Backend APIs (NextAuth, REST, SSE streaming, BullMQ queue dispatchers, ownership validation).
- Frontend UI (Next.js App Router, Tailwind CSS design system, responsive dashboards, interactive test interfaces).

3. Automated Testing:

- Agent authored Vitest unit suites for mathematics, RBAC data isolation, sliding-window rate limiters, and prompt injection defense.
- Playwright end-to-end tests covering complete multi-step user journeys.

4. Evaluation Harness:

- Agent wrote the programmatic evaluation framework (`src/lib/ai/eval/`) with test cases for factual grounding, insufficient evidence refusal, and prompt injection suppression.